

Expressions and Immutable Data

CS 350

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Overview

Programming in CS 350

Racket

Flit

THESE ARE YOUR
FATHER'S PARENTHESSES



ELEGANT
WEAPONS



FOR A MORE... CIVILIZED AGE.

Overview

Objectives

- To learn how to use Dr. Racket

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- To learn how to install and use `#lang flit`

Programming in CS 350

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- The Dr. Racket editor

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 - They are different from what we are doing. Beware!

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 - Can evaluate expressions while you're writing your code

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 - You can do a lot with very little

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- Data Types with Variants

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 - e.g. `if`, `lambda`, `type-case`

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 - NOT always a named function!

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Infix vs. Prefix

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- Languages like C++ and Python use “infix notation” for math

Function calls

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- In Flit, you write `( f x y z )`

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- In C++ you write `f(x, y, z)`
- In Flit, you write `(f x y z)`
- 99% of what you do in Flit is calling functions

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 - Hierarchical structure: expressions contain expressions contain expressions...
 - Parentheses make this very explicit

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( + 1 2 )
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 - Save Curly brackets for the second half of the class

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- JSON is basically just funny-looking parentheses

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(max 10 20)
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```
9
9.5
2/9
1e-12
20
1
```

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 - *Mutate* value of variable

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- Program is made up of *expressions*
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- Variables are *immutable*
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- Use functions and recursion to do all computation

Example: Imperative Conditionals

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```
if (someCondition)
  doThis
else
  doThat
```

- The if-statement doesn't have a value, but it can mutate (change) the state

Conditional Expressions

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;; another example  
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  (if (= 2 (+ 1 1))  
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- Boolean changes what the expression **is**, not what it does
 - e.g. Above, the 2nd if evaluates to 3 because the condition evaluates to true

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- Example:

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 - If you give a functional program the same inputs, you will always get the same outputs
- Example:
 - Math plus is different than Flit plus, needs numbers, not expressions

Imperative Breaks Equational Reasoning

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int x;  
x = 3;  
x = 4;
```

- Have $x = 3$ and $x = 4$, so surely $3 = 4$, right?

Equations for IF

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 - The two sides of the equation have the same run-time result
 - No matter what we choose for EXPR1 or EXPR2

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Functions

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(define (addOne [x : Number]) : Number  
  (+ x 1))
```

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;; Call the function we defined  
(addOne 10)
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```
(f arg1 arg2 arg3 ... argN)
```

- Each argument might be a variable or a literal, or a complex expression in brackets

Defining Multi-Argument Functions

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```
(define (isRemainder [x : Number]
                    [y : Number]
                    [remainder : Number])
  : Boolean
  (= remainder (modulo x y)))
(isRemainder 10 3 1)
(isRemainder 10 4 1)
```

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```
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General Form Defining Functions

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```
(define (functionName  
  [argName : argType]  
  ...  
  [argNameN : argTypeN]) : returnType  
  functionBody)
```

- Brackets around the function name, followed by arguments

General Form Defining Functions

- General form:

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- Later in the course we'll see another way of defining functions

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;; example
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(define (addone x)  
  (+ x 1))
```

```
;; in general
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- Kind of awkward, but just how it works

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 - We'll see this formally later

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 - Very useful for gradually building up a complex solution from previous smaller bits